1 Observation Model Physics: Antenna, Attitude, Relativity, and Range Corrections

This document covers the *deterministic* corrections of the observation model, other than geometry, the atmosphere, and the streamed SSR overlay. The Foundations observation equations (the undifferenced code equation and the undifferenced phase equation) introduce the satellite and receiver hardware biases B_s^i, b_s^i, $B_{r,s}$, $b_{r,s}$ only as symbols. The observation-geometry section gives the geometric range ρ_r^i and its linearisation. This document supplies every other deterministic term between the raw measurement and the modelled range:

- the antenna phase-centre geometry, and the satellite attitude frame it depends on;
- the carrier-phase wind-up;
- the relativistic clock and propagation corrections;
- the Sagnac (Earth-rotation) frame correction;
- the bias terms, realised through broadcast group delays and the precise `.BIA` product.

The streamed SSR overlay corrects the same terms in real time and is described in the SSR corrections document. The two mechanisms differ in frames, sign conventions, and gating, and the differences are noted where they matter.

`PppModel::computeBaseModeledValue` in `model/PppModel.cpp` assembles every term below into one modelled observable. Both the iono-free-pair and the uncombined observation paths call it. It adds the terms to the geometric range from `model/ModelGeometry.cpp`. For the code observable the sum is

$$\hat{P}_{r,s}^i = \rho_r^i \;-\; c\,\mathrm{d}t^i \;-\; c\,\delta t_{\mathrm{rel}} \;+\; T_r^i \;+\; A_{r,s}^i \;+\; B_{\mathrm{grp}} \;+\; S_r^i \;+\; d_{\mathrm{atm}} \;+\; d_{\mathrm{tide}}, \tag{1}$$

where T_r^i is the a priori slant hydrostatic delay (the wet delay and the gradients are estimated; the Atmosphere document), $A_{r,s}^i$ is the total antenna correction (Section 1.2), δt_{rel} the periodic relativistic clock term (Section 1.4), S_r^i the Shapiro delay (Section 1.5), B_{grp} the deterministic code/phase bias (Section 1.7), and d_{atm}, d_{tide} the loading/site-displacement terms documented in the site-displacement document. The Sagnac correction (Section 1.6) is part of ρ_r^i and is not added separately; the reason is given below. The signs in (1) follow the return statement of `PppModel::computeBaseModeledValue` (`model/PppModel.cpp`): the satellite clock and relativistic terms are *subtracted*, and the antenna, Shapiro, loading and tide terms are *added*.

1.1 Satellite attitude: the body frame

The satellite phase-centre offset, the phase-centre variation, and the carrier wind-up are all defined in the *spacecraft body frame*, so that frame is constructed first. The engine uses the IGS/ANTEX convention: a right-handed orthonormal triad $(\hat{\boldsymbol{x}}_i, \hat{\boldsymbol{y}}_i, \hat{\boldsymbol{k}}_i)$ with $+Z$ toward the Earth's centre (nadir / antenna boresight), $+X$ in the Sun–satellite–Earth plane pointing toward the Sun, and $+Y$ completing the triad along the solar-panel (cross-track) axis (`model/SatAttitude.cpp`, `model/Antenna.cpp`).

Nominal yaw-steering. With $\boldsymbol{r}^i$ the satellite ECEF position and $\hat{\boldsymbol{e}}_\odot = (\boldsymbol{r}_\odot - \boldsymbol{r}^i)/\|\boldsymbol{r}_\odot - \boldsymbol{r}^i\|$ the unit direction to the Sun, the nadir axis and the cross-track axis are

$$\hat{\boldsymbol{k}}_i = -\frac{\boldsymbol{r}^i}{\|\boldsymbol{r}^i\|}, \qquad \hat{\boldsymbol{y}}_i = \frac{\hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{e}}_\odot}{\|\hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{e}}_\odot\|}, \qquad \hat{\boldsymbol{x}}_i = \hat{\boldsymbol{y}}_i \times \hat{\boldsymbol{k}}_i, \tag{2}$$

in the operand order of `SatAttitude::compute` (`model/SatAttitude.cpp`). One model serves the satellite PCO and PCV and the wind-up: the antenna path (`Antenna::getAntennaCorrection`) and the wind-up (`GnssCorrections::windUpGeometry`) both take the frame from `SatAttitude`, so the two cannot disagree on the mode selection or on the sign conventions. Forming $\hat{\boldsymbol{y}}_i$ first and then $\hat{\boldsymbol{x}}_i = \hat{\boldsymbol{y}}_i \times \hat{\boldsymbol{k}}_i$ is the Gram–Schmidt orthogonalisation of the Sun direction against nadir. By the $\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c})$ identity,

$$\hat{\boldsymbol{x}}_i = \hat{\boldsymbol{y}}_i \times \hat{\boldsymbol{k}}_i = (\hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{e}}_\odot) \times \hat{\boldsymbol{k}}_i = \hat{\boldsymbol{e}}_\odot - \hat{\boldsymbol{k}}_i\,(\hat{\boldsymbol{k}}_i \cdot \hat{\boldsymbol{e}}_\odot), \tag{3}$$

so $\hat{\boldsymbol{x}}_i$ is the normalised component of $\hat{\boldsymbol{e}}_\odot$ orthogonal to nadir. The $+X$ axis therefore points toward the Sun as far as the nadir constraint allows, as the ANTEX convention requires. The body-frame matrix with these three columns,

$$\boldsymbol{F} = \big[\, \hat{\boldsymbol{x}}_i \;\; \hat{\boldsymbol{y}}_i \;\; \hat{\boldsymbol{k}}_i \,\big], \tag{4}$$

is orthonormal, so its inverse is its transpose. This is used below to rotate a body-frame offset into ECEF.

Sun elevation and the mode switch. Non-nominal attitude (eclipse seasons, the noon and midnight yaw turns) depends on the Sun's elevation above the orbital plane, β. The orbit plane is inertial, so the orbit angular-momentum (normal) unit vector $\hat{\boldsymbol{n}} = (\boldsymbol{r}^i \times \dot{\boldsymbol{r}}^i_{\mathrm{I}})/\|\boldsymbol{r}^i \times \dot{\boldsymbol{r}}^i_{\mathrm{I}}\|$ is formed from the inertial velocity $\dot{\boldsymbol{r}}^i_{\mathrm{I}} = \dot{\boldsymbol{r}}^i_{\mathrm{ECEF}} + \boldsymbol{\omega}_\oplus \times \boldsymbol{r}^i$ (`SatAttitude::inertialVelocity`, the inverse of the product-evaluation velocity transform),

$$\boxed{\beta = \arcsin\big(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{e}}_\odot\big).} \tag{5}$$

Forming $\boldsymbol{r}^i \times \dot{\boldsymbol{r}}^i_{\mathrm{ECEF}}$ instead removes $\boldsymbol{r}^i \times (\boldsymbol{\omega}_\oplus \times \boldsymbol{r}^i)$ from the normal, a vector of 29 to 59 % of $\|\boldsymbol{r}^i \times \dot{\boldsymbol{r}}^i_{\mathrm{I}}\|$ at GPS and Galileo orbits, which tilts the normal by tens of degrees. The conversion applies when `estimation.enable_sun_frame_correction` is set (the Sun and Moon section of the Coordinate Frames document); with it off, the ECEF velocity is used as before, bit for bit. The noon-turn model below always converts. The angle is signed by the projection of the Sun direction onto the $+(\boldsymbol{r}^i \times \dot{\boldsymbol{r}}^i)$ normal (`SatAttitude::computeBeta`). With the conversion on, it agrees with an independent computation (SP3 positions rotated to the inertial frame and an almanac Sun, `scripts/attitude_beta_check.py`) to 0.01°, with no sign mismatch, on seven GPS satellites over a day. The mode selector (`SatAttitude::determineMode`) keeps GPS, GLONASS and Galileo in yaw-steering unconditionally, and switches BeiDou and QZSS to orbit-normal attitude below a constellation-specific β threshold (4.0° BeiDou, 20.0° QZSS, `model/SatAttitude.cpp`). The threshold compares the signed β, as NGS M-PAGES does. In orbit-normal mode the $+Y$ axis is pinned to the orbit normal, $\hat{\boldsymbol{y}}_i = \hat{\boldsymbol{n}}$, and the triad is re-orthogonalised $\hat{\boldsymbol{x}}_i = \hat{\boldsymbol{y}}_i \times \hat{\boldsymbol{k}}_i$, $\hat{\boldsymbol{y}}_i = \hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{x}}_i$. The axes are those of the positive orbit normal in every orbit-normal case; the QZSS label `OrbitNormalNegative` serves traces only. The legacy default path also forces a hard-coded list of BeiDou-2 GEO PRNs (C01–C05, C59, C60) into orbit-normal mode whatever the value of β. The β-driven table is active only when `processing.enable_orbit_normal_attitude` is set (default off). With the flag off the result is bit-identical to the earlier inline frame (`model/Antenna.cpp`, `model/SatAttitude.cpp`). The

wind-up takes the same frame but keeps only the BeiDou-2 GEO orbit-normal rule and yaw-steers every other satellite whatever that flag says; the flag moves the PCO frame alone. Without a Sun direction the frame is built from the velocity direction about the nadir axis (`SatAttitude::velocityFrame`), and a Sun collinear with nadir falls back to $\hat{\boldsymbol{y}}_i \propto \hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{v}}$.

Nominal yaw steering for Galileo and the other systems. GPS, GLONASS and Galileo satellites carry the nominal frame (2) at every β. The model has no manoeuvre law for Galileo or GLONASS, no eclipse or midnight turn for any system, and no yaw bias. A Galileo satellite in the Earth's shadow is therefore excluded upstream when `processing.exclude_eclipsing_satellites` is set (default off, on in the reference configurations), and the nominal frame is used otherwise.

GPS noon turn. Near the orbit noon at small $|\beta|$ the nominal yaw would need a rate above what the attitude controller of a GPS Block IIF or IIR satellite delivers. The satellite then yaws at its maximum rate and rejoins the nominal yaw afterwards [**kouba2009simplified**]. With μ the orbit angle from the orbit midnight (0 at midnight, π at noon) and $\dot{\mu}$ the mean orbit rate 0.00836°/s, the nominal yaw about the nadir axis, measured from the velocity direction, is

$$\psi_{\mathrm{nom}} = \mathrm{atan2}\big(-\tan\beta,\ \sin\mu\big), \tag{6}$$

and its rate peaks at noon at $\dot{\mu}/|\tan\beta|$. With $\dot{\psi}_{\max}$ the controller limit (0.11°/s for Block IIF, 0.20°/s for Block IIR, IIR-A, IIR-B and IIR-M, chosen from the block named in the ATX satellite type), no turn occurs when $|\beta| \geq \beta_0 = \arctan(\dot{\mu}/\dot{\psi}_{\max})$, that is 4.35° and 2.39°. Below β_0 the satellite follows ψ_{nom} until the nominal rate reaches $\dot{\psi}_{\max}$, at the orbit angle x_s before noon with $c = \cos x_s$ the positive root of

$$\dot{\psi}_{\max}\, c^2 + \dot{\mu}\,|\tan\beta|\, c - \dot{\psi}_{\max}\,(1 + \tan^2\beta) = 0, \tag{7}$$

then yaws at $\dot{\psi}_{\max}$ against the sign of β until the ramp meets ψ_{nom} again. Inside the turn the body X and Y axes are rotated about nadir by the difference $\Delta = \psi_{\mathrm{act}} - \psi_{\mathrm{nom}}$ (wrapped to $\pm\pi$):

$$\hat{\boldsymbol{x}}'_i = \cos\Delta\ \hat{\boldsymbol{x}}_i + \sin\Delta\,(\hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{x}}_i), \qquad \hat{\boldsymbol{y}}'_i = \hat{\boldsymbol{k}}_i \times \hat{\boldsymbol{x}}'_i, \tag{8}$$

and outside the turn the frame is returned bit for bit (`SatAttitude::noonTurnYaw`, `applyNoonTurn`, `noonTurnLaw`). The same rotated frame feeds the satellite PCO and PCV (`Antenna::getAntennaCorrection`) and the wind-up dipole (14), so the two stay on one attitude. The model applies only to GPS Block IIF and IIR satellites and only when `processing.enable_noon_turn_attitude` is set (default off; on in the real-data reference configuration, listed as a documented exception in the simulated-network one because the simulated data carry no noon-turn law). The model is described in `docs/refound/NTA_noon_turn_attitude_2026-10-16.md`.

Eclipse gating. The body frame above assumes a known yaw angle. That assumption fails during Earth-shadow passes. The actual yaw during eclipse and the recovery after it depend on the satellite block and are not modelled here, so the antenna PCO/PCV and the wind-up are unreliable for an eclipsing satellite. When the user opts in (`processing.exclude_eclipsing_satellites`), the satellite is dropped at the geometry stage by a *cylindrical umbra* test (`Planetary::isSatelliteEclipsing`, `model/Planetary.cpp`, called from `model/ModelGeometry.cpp`). With $\hat{\boldsymbol{u}}_\odot = \boldsymbol{r}_\odot/\|\boldsymbol{r}_\odot\|$ the Earth-to-Sun direction, the satellite's along-axis and perpendicular distances are $s_\parallel = \boldsymbol{r}^i \cdot \hat{\boldsymbol{u}}_\odot$ and $\boldsymbol{s}_\perp = \boldsymbol{r}^i - s_\parallel\,\hat{\boldsymbol{u}}_\odot$, and the predicate is

$$\text{eclipsed} \iff \big(s_\parallel < 0\big)\ \wedge\ \big(\|\boldsymbol{s}_\perp\| < \kappa\, R_\oplus\big), \tag{9}$$

i.e. the satellite is on the anti-Sun side *and* inside the shadow cylinder of radius $\kappa R_\oplus$, with $R_\oplus = 6\,371\,230\,\mathrm{m}$ (`common/Constants.hpp`) and κ the user shadow scale (`processing.eclipse_shadow_scale`, default unity; cited rather than reproduced as a tuning knob). This is a conservative geometric umbra test. It does not model the penumbra.

1.2 Antenna phase-centre corrections

The range is measured between the satellite and receiver *antenna phase centres*. The estimated states refer to the satellite centre of mass and the receiver antenna reference point (ARP). The position state must be the ARP because the ANTEX receiver PCO/PCV are referenced to it. The antenna correction $A^i_{r,s}$ connects the two, as the sum of four geometric terms and the wind-up:

$$\boxed{A^i_{r,s} \;=\; -\,\delta^i_{\mathrm{PCO}} \;+\; \delta^i_{\mathrm{PCV}} \;+\; \delta^r_{\mathrm{PCO}} \;+\; \delta^r_{\mathrm{PCV}} \;+\; \varphi_{\mathrm{wu}},} \tag{10}$$

as accumulated in `Antenna::getAntennaCorrection` (`model/Antenna.cpp`). The line of sight used in this section is the *satellite-to-receiver* unit vector, $\boldsymbol{e} = (\boldsymbol{r}_r - \boldsymbol{r}^i)/\|\boldsymbol{r}_r - \boldsymbol{r}^i\|$ (`Antenna.cpp`). It is the negative of the receiver-to-satellite $\boldsymbol{e}^i_{\hat{r}}$ of the geometry section (its line-of-sight definition). The sign of each term below follows from this choice.

Which coordinate is the ARP. Every position the engine derives from a RINEX *header* is an ARP position. The marker coordinate is reduced to the ARP through the header eccentricity before it seeds the state (`Coordinates::markerToArp`, packaged as `RinexParser::getArpPositionXyz`; the marker/ARP section of the Coordinate Frames document). A coordinate supplied directly in the configuration is *not* reduced. It is used as given, on the policy that the user owns its reference point. This matters in the network estimator. Its station coordinates come from the configuration (`StationConfig::position_ecef`) and are held fixed, with their position columns dropped from the normal equations. A monument coordinate there is not absorbed by a position state. The offset goes into the residuals, and from there into the estimated clocks, biases, and ionosphere. `network.station_marker_to_arp` (default off, on in both reference configurations) reduces each roster coordinate once at setup; a SIF coordinate is reduced through the same function (the marker/ARP section of the Coordinate Frames document, the engine section of the Reference Frames document).

Satellite PCO (subtracted). The satellite phase-centre offset $\boldsymbol{p}^i_{\mathrm{body}}$ is published in the body frame (millimetres in ANTEX). It is rotated to ECEF by the body-frame matrix; because $\boldsymbol{F}$ collects the body axes as *columns*, $\boldsymbol{F}$ itself maps body $\to$ ECEF (and the orthonormal $\boldsymbol{F}^{\mathrm{T}} = \boldsymbol{F}^{-1}$ maps ECEF $\to$ body). It is then projected onto the LOS:

$$\delta^i_{\mathrm{PCO}} = \left(\boldsymbol{F}\,\boldsymbol{p}^i_{\mathrm{body}}\right) \cdot \boldsymbol{e}. \tag{11}$$

The antenna phase centre is *nearer the receiver* than the centre of mass, so the true range is shorter than the CoM range and the correction is *subtracted* in (10) (`correction -= sat_pco_corr`, `model/Antenna.cpp`). There is one real-time exception. When an applied SSR orbit correction is referenced to the antenna phase centre, the corrected orbit already points at the phase centre. The `skip_satellite_pco` path of `getAntennaCorrection` then drops the satellite-PCO term so that it is not counted twice. The path is gated by `realtime.honor_ssr_reference_point` and is off by default. The SSR reference-point convention is documented in the SSR corrections document.

Satellite PCV (added). The satellite phase-centre variation is a function of the *nadir angle*, the angle between the boresight $\hat{\boldsymbol{k}}_i$ and the line of sight, and of the azimuth in the body frame. With $\cos\theta_{\mathrm{nad}} = \hat{\boldsymbol{k}}_i \cdot \boldsymbol{e}$,

$$\theta_{\mathrm{nad}} = \arccos\big(\hat{\boldsymbol{k}}_i \cdot \boldsymbol{e}\big), \qquad \alpha_{\mathrm{body}} = \mathrm{atan2}\big((\boldsymbol{F}^{\mathrm{T}}\,\boldsymbol{e})_y,\,(\boldsymbol{F}^{\mathrm{T}}\,\boldsymbol{e})_x\big), \tag{12}$$

and the PCV value is read by bilinear interpolation of the ANTEX grid over (zenith/nadir ring, azimuth sector), the look-up argument being $\theta_{\mathrm{nad}} - \texttt{zen1}$ in grid units (`Antenna::interpolatePcv`, `model/Antenna.cpp`). The satellite PCV is *added* to the range correction.

Receiver PCO (added). The receiver PCO is published as a North/East/Up (NEU) triplet relative to the ARP. It is reordered to ENU, rotated to ECEF by the local ENU→ECEF rotation $\boldsymbol{R}_{\mathrm{enu}}(\phi,\lambda)$ at the receiver geodetic latitude/longitude (`Coordinates::enuToEcefMatrix`), and projected on the LOS:

$$\delta^{r}_{\mathrm{PCO}} = \Big(\boldsymbol{R}_{\mathrm{enu}}(\phi,\lambda)\,\big[\,p_E\;\;p_N\;\;p_U\,\big]^{\mathrm{T}}\Big)\cdot\boldsymbol{e}, \qquad \big[\,p_E\;p_N\;p_U\,\big] = \big[\,p_{\mathrm{NEU},2}\;\;p_{\mathrm{NEU},1}\;\;p_{\mathrm{NEU},3}\,\big], \tag{13}$$

where the index swap $(N,E,U)\to(E,N,U)$ is explicit in `model/Antenna.cpp`. It is *added* to the correction.

Receiver PCV (added). The receiver PCV is indexed by *zenith angle* $z = 90^\circ - \mathrm{el}$ (and, optionally, azimuth corrected for the antenna mounting orientation when `enableRxAntennaOrientation()` is set), with the same bilinear interpolation over $z - \texttt{zen1}$ (`model/Antenna.cpp`). It is *added* to the correction.

ATX frequency fallback. ANTEX entries often omit the newest carriers: L5/E5a/B2a on older spacecraft, and many receiver antennas published before triple-frequency tracking. Setting the antenna correction to zero would leave a vertical bias of several centimetres. The engine instead substitutes the *nearest carrier* when the requested ANTEX frequency key is absent (`model_corrections.enable_atx_frequency_fallback`, default on; `pickAtxFrequencyKey`, `model/Antenna.cpp`). The chains are ordered by carrier-frequency proximity, e.g. for GPS `G05` → {`G02`, `G01`} and for Galileo `E05` → {`E08`, `E07`, `E06`, `E01`}. The PCO of one physical antenna typically changes by less than a centimetre between frequencies. The PCO and PCV chains are searched *independently*, because an antenna may have a PCO but no PCV at a given key.

For the *satellite* antenna only the chain within the same system is allowed; another constellation's spacecraft entry would be meaningless. For the *receiver* antenna a *cross-system* chain to the nearest GPS band is allowed after the same-system chain is exhausted, and only when `model_corrections.enable_receiver_cross_system_atx_fallback` is set (default off; on in both reference configurations, where the Galileo E1 and E5a signals take the GPS L1 and L5 receiver entries; `pickReceiverAtxFrequencyKey`). This is justified because the receiver phase pattern depends on frequency, not on constellation. The important mappings are exact carrier matches (`E01` → `G01`, `E05` → `G05` at 1575.42 and 1176.45 MHz). The rest are approximations 14 to 51 MHz apart (`model/Antenna.cpp`). Each substitution sets a coverage flag and writes a log line. The same fallback policy is used for biases: substitute and report, and do not exclude the measurement.

1.3 Carrier-phase wind-up

A right-hand circularly polarised GNSS signal gains a phase shift equal to the relative rotation of the transmitting and receiving antenna dipoles about the line of sight [2]. This wind-up is the

same in cycles on every signal, so as a range it is proportional to wavelength. It is computed in `GnssCorrections::computePhaseWindUp` (`corrections/GnssCorrections.cpp`). With $\boldsymbol{k}$ the unit line-of-sight, $(\hat{\boldsymbol{x}}_i, \hat{\boldsymbol{y}}_i)$ the satellite body $\pm X, +Y$ axes and $(\boldsymbol{x}_r, \boldsymbol{y}_r)$ the receiver local North/East axes, the *effective dipoles* are

$$\boldsymbol{D}_i = \hat{\boldsymbol{x}}_i - \boldsymbol{k}\,(\boldsymbol{k}\cdot\hat{\boldsymbol{x}}_i) - \boldsymbol{k}\times\hat{\boldsymbol{y}}_i, \qquad \boldsymbol{D}_r = \boldsymbol{x}_r - \boldsymbol{k}\,(\boldsymbol{k}\cdot\boldsymbol{x}_r) - \boldsymbol{k}\times\boldsymbol{y}_r, \tag{14}$$

which is the standard Wu construction. Each dipole is the antenna X axis projected into the plane orthogonal to $\boldsymbol{k}$, *minus* the $\boldsymbol{k}\times Y$ quadrature term (`corrections/GnssCorrections.cpp`).

The sign of the quadrature term is fixed by the antenna axes. It is $-\operatorname{sgn}\big((\boldsymbol{x}\times\boldsymbol{y})\cdot\boldsymbol{k}\big)$, which is negative when the antenna's third axis points along $+\boldsymbol{k}$. That is the case for both triads here: the satellite's $\boldsymbol{e}_z$ is the geocentric nadir, and the receiver's implied third axis is North × East = Down. The wrong sign gives $|\boldsymbol{D}| = 1 - \cos\varepsilon$, a dipole that *vanishes on boresight*. RTKLIB's $(-,+)$ pairing is stated with $\boldsymbol{y}_r = -$East and is the same construction; with the +East convention used here it becomes $(-,-)$. The construction with both signs negative [**wu1993effects**] replaced a $(+,+)$ pairing on 2026-08-21 (`docs/SPEC_phase_windup_sign.md`); the revision is declared as `nominal-wu93` (see the paragraph on the shared convention below).

The receiver North and East axes are built from the geodetic latitude ϕ and longitude λ, $\boldsymbol{x}_r = (-\sin\phi\cos\lambda,\ -\sin\phi\sin\lambda,\ \cos\phi)$ and $\boldsymbol{y}_r = (-\sin\lambda,\ \cos\lambda,\ 0)$ (`GnssCorrections.cpp`). The signed wind-up angle is the atan2 of the cross and dot products of the unit dipoles. This covers the full range $(-\pi, \pi]$ and avoids the loss of precision of an arccos near 0 and π:

$$\boxed{\Delta\varphi = \operatorname{atan2}\Big(\boldsymbol{k}\cdot\big(\hat{\boldsymbol{D}}_i\times\hat{\boldsymbol{D}}_r\big),\ \hat{\boldsymbol{D}}_i\cdot\hat{\boldsymbol{D}}_r\Big), \qquad \varphi_{\mathrm{wu}} = \Delta\varphi\,\frac{\lambda_s}{2\pi},} \tag{15}$$

with $\hat{\boldsymbol{D}}_i, \hat{\boldsymbol{D}}_r$ the normalised dipoles and λ_s the signal wavelength (`corrections/GnssCorrections.cpp`); the $\boldsymbol{k}\cdot(\hat{\boldsymbol{D}}_i\times\hat{\boldsymbol{D}}_r)$ argument fixes the sign of the rotation about the line of sight. The result enters (10) as φ_{wu} on phase observables only (the code branch passes zero wavelength).

Fractional vs. accumulated modes. The atan2 returns the wind-up wrapped to one cycle. The *fractional* mode (`estimation.accumulate_phase_windup` false, the loader default) uses this wrapped angle directly. The float ambiguity $N^i_{r,s}$ absorbs any whole-cycle part, so the wrapping does no harm to a float solution, but each crossing of $\pm\pi$ steps the applied correction by one cycle on every signal of the satellite. Integer ambiguity resolution needs the ambiguity to stay a constant integer across the arc. For that the engine offers an *accumulated* (unwrapped) mode, which unwraps each epoch relative to the previous accumulated angle A_{k-1} and restarts from the raw angle $\Delta\varphi_k$ otherwise:

$$A_k = \begin{cases} A_{k-1} + \operatorname{wrap}_{2\pi}\big(\Delta\varphi_k - A_{k-1}\big), & 0 < t_k - t_{k-1} < T_{\mathrm{wu}}, \\ \Delta\varphi_k \quad \text{(restart)}, & \text{otherwise}, \end{cases} \qquad \operatorname{wrap}_{2\pi}(x) = x - 2\pi\,\operatorname{round}\!\big(\tfrac{x}{2\pi}\big), \tag{16}$$

with T_{wu} the continuity horizon `estimation.phase_windup_max_gap_s` (default 300 s). The accumulator keeps one angle and one epoch per satellite; a repeated call within one epoch (once per signal and per rebuild pass) reuses the accumulated value and is not a restart. A larger gap, or a backward time step, restarts the accumulator (`corrections/GnssCorrections.cpp`, `computePhaseWindUp`). By (16), $A_k - \Delta\varphi_k \in 2\pi\mathbb{Z}$, so a restart steps the modelled phase of every signal of the satellite by the same whole number of cycles. On a single receiver the step cannot bias the solution, because an absent satellite's ambiguity is pruned and re-seeded on re-acquisition. Each restart is recorded as a (satellite, gap, epoch) entry, which the network's event response reads (the wind-up section of the Network Event Responses document). Both reference configurations accumulate.

Gap bridge. `estimation.phase_windup_gap_bridge` (default off; it needs the accumulated mode, and the loader switches it off with a warning otherwise) replaces the restart for a gap g with $T_{\mathrm{wu}} \le g < T_{\mathrm{br}}$, where T_{br} is `estimation.phase_windup_bridge_max_gap_s` (default 900 s, the horizon of the network's phase-leveling carry). The angle is walked across the gap through virtual epochs. Cut the gap into $n = \lceil g/\Delta \rceil$ equal steps, with $\Delta =$ `estimation.phase_windup_bridge_step_s` (default 30 s, accepted in $(0, 120]$). The raw angle $\theta(\cdot)$ at a virtual epoch comes from the same routine as the epoch's own angle, with the satellite state taken from the orbit products and the receiver where it is now (`Geometry::getLookBackState`). Each step is evaluated at its middle as well, and each half step is wrapped to half a cycle either way:

$$W = \sum_{i=1}^{n} \Big[\mathrm{wrap}_{2\pi}\big(\theta(t_{i-\frac{1}{2}}) - \theta(t_{i-1})\big) + \mathrm{wrap}_{2\pi}\big(\theta(t_i) - \theta(t_{i-\frac{1}{2}})\big) \Big], \tag{17}$$

$$N = \mathrm{round}\Big(\frac{A_{k-1} + W - \Delta\varphi_k}{2\pi}\Big), \tag{18}$$

with $t_i = t_{k-1} + i\,g/n$. The bridged angle is $A_k = \Delta\varphi_k + 2\pi N$: the fractional part is the epoch's own, and the walk supplies only the integer, so an error of the virtual geometry matters only if it moves the sum by a quarter of a cycle. The walk is refused, and the restart happens as without the bridge, when a virtual epoch has no satellite state or a degenerate geometry; when the satellite is in the Earth's umbra there (`Planetary::isSatelliteEclipsing`, with `processing.eclipse_shadow_scale`); when it is in a GPS noon-turn window, widened by one step of orbit motion (`SatAttitude::inNoonTurnWindow`); when a half step moves the angle by a quarter of a cycle or more; and when $|A_{k-1} + W - \Delta\varphi_k - 2\pi N|$ exceeds a quarter of a cycle (`WindUpBridge::walk`). Each attempt is counted by outcome (`WindUpBridge::Counters`), and the first attempts of a receiver are kept with their satellite, epoch and gap. The setting is UNVALIDATED in `docs/CONFIG_promotion_ledger.json`: in a six-hour real-data case the restarts fell from 139 to 91 and the refloated arcs from 116 to 23 with no change in the test station beyond 0.3 mm (`docs/refound/WGB_windup_gap_bridge_2026-10-17.md`). The simulator injects wind-up through the same function, so no simulated data can grade it.

One convention for the network, the user and the simulator. The wind-up angle depends on the satellite attitude model, the sign rule of the dipoles and the accumulation mode. A mismatch between the network and a user is absorbed by the user's ambiguities and is invisible in the variance factor. The construction is declared by the compile-time constant `kWindUpYawModelId` = `"nominal-wu93"` (`network/solve/Sbasis.hpp`), written into the contract line of every emitted product, and a product that declares another value is refused by `NetworkProductReader::checkContract`. The bare token `"nominal"` names the earlier $(+,+)$ construction, whose products carry $2\varphi_{\mathrm{wu}}$ in their satellite clocks and phase biases. The suffix names a revision of the wind-up construction. The token does not record the accumulation mode or the noon-turn setting, so those are held by configuration. The network, the users and the simulator evaluate the same function: the simulator injects its wind-up through `GnssCorrections::computePhaseWindUp` in accumulated mode (`src/sim/EphemerisProvider.cpp`), so the simulated network, the network estimator and the user share one model.

1.4 Periodic relativistic clock correction

A clock on an eccentric orbit runs at a varying rate relative to a clock on the geoid. The broadcast or precise clock polynomial absorbs the constant and secular parts. The *periodic* part, caused by

orbital eccentricity, must be applied separately. The standard expression [1], in `GnssCorrections::computeRelativisticCorrection` (`corrections/GnssCorrections.cpp`), is

$$\boxed{\delta t_{\mathrm{rel}} \;=\; -\,\frac{2\,(\boldsymbol{r}^i \cdot \dot{\boldsymbol{r}}^i)}{c^2} \quad [\mathrm{s}], \qquad \text{range term } = -\,c\,\delta t_{\mathrm{rel}},} \tag{19}$$

applied to the range by the $-c\,\delta t_{\mathrm{rel}}$ term of the assembly (1) (`model/PppModel.cpp`). Two points need stating.

First, the dot product $\boldsymbol{r}^i \cdot \dot{\boldsymbol{r}}^i$ is the same whether the position and velocity are expressed in the inertial frame or in the rotating ECEF frame. The ECEF velocity is $\dot{\boldsymbol{r}}^i_{\mathrm{ecef}} = \dot{\boldsymbol{r}}^i_{\mathrm{inertial}} - \boldsymbol{\omega}_\oplus \times \boldsymbol{r}^i$, and $\boldsymbol{r}^i \cdot (\boldsymbol{\omega}_\oplus \times \boldsymbol{r}^i) = 0$ because the cross product is orthogonal to $\boldsymbol{r}^i$. The ECEF velocity from the broadcast ephemeris can therefore be used directly.

Second, the periodic term is computed here because the IGS precise CLK product, by convention, contains only the constant and drift parts of the clock. Applying (19) supplies the periodic part consistently for both precise and broadcast clock sources.

1.5 Shapiro (gravitational-delay) correction

The Earth's gravitational potential slows and bends the signal. The resulting Shapiro delay is a few centimetres [1]. With $r_i = \|\boldsymbol{r}^i\|$, $r_r = \|\boldsymbol{r}_r\|$ the geocentric radii and $r_{ir} = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|$ the satellite–receiver distance, the delay (in metres) is

$$\boxed{S^i_r \;=\; \frac{2\,GM}{c^2}\,\ln\frac{r_i + r_r + r_{ir}}{r_i + r_r - r_{ir}},} \tag{20}$$

with $GM = 3.986005 \times 10^{14}\,\mathrm{m}^3/\mathrm{s}^2$ (`Constants::GRAVITATIONAL_CONSTANT`, `common/Constants.hpp`) and the formula as coded in `GnssCorrections.cpp`. Since $r_{ir} < r_i + r_r$ the logarithm is positive, so $S^i_r > 0$ and the term is *added* to the modelled range in (1) (`model/PppModel.cpp`). It is ∼1–2 cm for a typical MEO geometry.

1.6 Sagnac (Earth-rotation) correction

The pseudorange relates the satellite position at *signal emission* to the receiver position at *signal reception*, and both must be expressed in one ECEF frame. The ECEF frame rotates with the Earth during the ∼70 ms transit, so the emission-time satellite coordinates are rotated into the reception-time frame. The engine applies the full rotation, not a scalar approximation. With the geometric time-of-flight $\tau = \|\boldsymbol{r}^i_{\mathrm{em}} - \boldsymbol{r}_r\|/c$ and the Earth rotation rate $\omega_\oplus = 7.2921151467 \times 10^{-5}\,\mathrm{rad/s}$ (`common/Constants.hpp`), the emission-time position $\boldsymbol{r}^i_{\mathrm{em}}$ is rotated by $\theta = \omega_\oplus \tau$ about the Z axis,

$$\boxed{\boldsymbol{r}^i = \boldsymbol{R}_z(\omega_\oplus \tau)\,\boldsymbol{r}^i_{\mathrm{em}}, \qquad \boldsymbol{R}_z(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix},} \tag{21}$$

after which $\rho^i_r = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|$ and the line of sight are computed from the rotated $\boldsymbol{r}^i$ (`model/ModelGeometry.cpp`, the `rotateAndComputeTopo` lambda). The Sagnac correction is therefore *part of the geometric range*, which is why (1) has no separate Sagnac term. An equivalent scalar form, $(\omega_\oplus/c)\,(x_i y_r - y_i x_r)$, exists in the code as a *diagnostic* only (`GnssCorrections::computeSagnacCorrection`, `corrections/GnssCorrections.cpp`). It is the first-order expansion of the rotation and is not used for modelling. The Orbit and Clock Product Evaluation document

refers to this section for the Sagnac rotation. The same rotation $\boldsymbol{R}_z(\omega(t_{\text{node}} - t_{\text{query}}))$ is used at SP3 interpolation time to rotate ephemeris nodes into the query-epoch frame (`model/ModelGeometry.cpp`), for numerical conditioning, as described with the products.

1.7 Group delays and inter-frequency biases

Two products supply deterministic values for the bias terms of the Foundations observation equations: the satellite code and phase biases B_s^i, b_s^i and the receiver biases $B_{r,s}$, $b_{r,s}$. This section relates the products to those symbols.

Signals of the reference configurations. Both reference configurations (`config/golden/golden_field_refound.json` and `golden_sim_refound.json`) run GPS L1/L2 and Galileo E1/E5a in the dual-frequency mode (`processing.frequency_mode dual`), with one signal per band chosen by the priority rule. The Galileo priority list is `C1C`, `C1X`, `L1C`, `L1X`, `C5Q`, `C5X`, `L5Q`, `L5X`: each band is admitted in both tracking families, the pilot-only signal (`C1C`, `C5Q`) and the combined data-plus-pilot signal (`C1X`, `C5X`). The clock datum across the two families belongs to the network (the S-basis section of the Network document).

Broadcast group delay. The broadcast navigation message carries a per-system inter-frequency group delay (GPS T_{GD}, Galileo BGD, BeiDou TGD). It refers each signal's code hardware delay to the ionosphere-free combination used to generate the broadcast clock. The observation model does not apply it as a correction: the broadcast clock of the product-evaluation document omits it by convention, so that it matches the precise-clock baseline, and the ionosphere-free combination is free of it. The satellite code bias of an uncombined row is the precise-product value below or an estimated state. The delay is used by the network as the level of the satellite geometry-free code bias. For GPS the broadcast clock is the L1/L2 ionosphere-free clock, so $d_1 = c\,T_{\text{GD}}$ and $d_2 = \gamma\, c\, T_{\text{GD}}$ with $\gamma = f_1^2/f_2^2$, which makes the ionosphere-free combination of the two delays exactly zero and the geometry-free combination exactly $c\,T_{\text{GD}}$:

$$\frac{f_1^2 d_1 - f_2^2 d_2}{f_1^2 - f_2^2} = 0, \qquad \frac{d_2 - d_1}{\mu_2 - \mu_1} = c\,T_{\text{GD}}. \tag{22}$$

With `network.sat_gf_datum` = `broadcast_tgd` (both reference configurations) one pin row per satellite fixes that combination, over the pair of the ionosphere-free pin, at $c\,T_{\text{GD}}$ for GPS and at $c\,\text{BGD}(\text{E1}, \text{E5a})$ for Galileo, and the pin is carried in time (`network.sat_gf_datum_time_linked`). The datum is developed in the S-basis section of the Network document.

Precise OSB / DCB biases. When a precise bias product is loaded, the deterministic correction is the observable-specific bias (OSB) / differential code bias (DCB) entry for the exact (i,s) pair, time-windowed against its validity interval and returned by `GnssCorrections::getBiasCorrection` (`corrections/GnssCorrections.cpp`). This single scalar realises B_s^i (code) or b_s^i (phase) for that signal:

$$B_{\text{grp}} = \text{bias}(i, s, t), \qquad t_{\text{start}} \le t \le t_{\text{end}}, \tag{23}$$

with the application gated by `model_corrections.enable_code_phase_bias` (`model/PppModel.cpp`; default on, and off in both reference configurations, where the network estimates the satellite code biases itself). The bias index (`GnssCorrections::buildBiasIndex`) excludes two kinds of entry.

Zero-valued code entries, such as the MGEX placeholder `C2I`/`C6I` for BeiDou, are treated as *absent*. Otherwise the model would apply a null correction and also suppress the estimated bias state for that (i,s), leaving a real delay of tens of metres unmodelled.

GLONASS code entries are also treated as missing. The FDMA per-channel receiver delay cannot be represented by one value per (i,s), so those signals use the estimated-state path (`corrections/GnssCorrections.cpp`).

Estimated vs. deterministic biases. The estimator handles whatever the deterministic bias does not correct: mainly the receiver hardware delay $B_{r,s}$, $b_{r,s}$, and any satellite delay that no product covers. The receiver biases are common to all satellites on a signal. They are absorbed by the receiver clock and inter-system bias states, or cancel under between-satellite differencing (the differencing section). The per-(i,s) satellite code bias may be *estimated* as a state when no product covers it (`estimation.estimate_sat_biases`, `addSatelliteCodeBias` of `measurement/ObservationEquation.cpp`). The network estimator estimates the satellite code biases of both reference configurations through its own columns (`network.estimate_sat_code_bias`). The state-allocation gate that prevents estimating a bias the products already supply, hasBIA(i, s) from the static product, OR'd with hasSsrCB(i, s) from the live stream, is documented with the SSR overlay (the satellite-bias gate of the SSR corrections document); the present document supplies only the *deterministic* half of that OR.

Relation to the SSR overlay. The streamed SSR code/phase OSB (the SSR corrections document) refines the *same* B^i_s, b^i_s terms in real time, as a known term of the observation equation (`addSsrBias` in `measurement/ObservationEquation.cpp`, with the sign constant `kSatBiasSign`) alongside the deterministic B_{grp} here. The network estimator takes no SSR bias term, because it estimates the biases as states and importing the same quantity would count it twice. The SSR bias is a per-(i,s) scalar like (23). The SSR *orbit* delta is different: it is expressed in a radial/along/cross (RAC) frame (the RAC frame of the SSR document), which is *not* the satellite body frame (4) of this document. The RAC frame is built from the broadcast position and velocity to express an orbit correction. The body frame is built from the Sun direction to orient the antenna. The two frames differ in construction and in purpose.

1.8 Provenance and assembly

All the deterministic corrections are combined in `model/PppModel.cpp`. `computeBaseModeledValue`, shared by the iono-free-pair and uncombined paths, returns the sum (1). It calls `model/ModelGeometry.cpp` for the Sagnac-rotated geometric range (21), `model/Antenna.cpp` for the total antenna correction (10) (which in turn calls `model/SatAttitude.cpp` for the body frame and `corrections/GnssCorrections.cpp` for the wind-up), and `corrections/GnssCorrections.cpp` for the relativistic (19), Shapiro (20) and bias (23) terms. The Sun and Moon ephemeris feeding the attitude frame (2), the Sun-elevation (5), the eclipse predicate (9) and the wind-up (14) is produced by `model/Planetary.cpp` (an analytic RTKLIB/Meeus series rotated to ECEF through the IAU 1976/1980 precession-nutation chain, cached per epoch) and is documented in the Coordinate Frames and Time Systems document. Every term is a fixed physical model. The user-facing knobs are enable toggles and the eclipse-shadow scale; they are cited rather than reproduced because they set *whether* a correction is applied, never its functional form.

References

[1] J. Sanz Subirana, J. M. Juan Zornoza, and M. Hernández-Pajares. *GNSS Data Processing, Volume I: Fundamentals and Algorithms*. ESA TM-23/I. Noordwijk, The Netherlands: ESA Communications, 2013.

[2] G. Wübbena. "Software developments for geodetic positioning with GPS using TI 4100 code and carrier measurements". In: *Proceedings of the First International Symposium on Precise Positioning with the Global Positioning System*. Rockville, MD, USA, 1985, pp. 403–412.